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The student named above has successfully defended the thesis or dissertation. By signing below, the committee members approve the thesis or dissertation, and certify that all required corrections have been proposed.

Chair (Printed Name/ Signature): Stephanie Stots

Committee Members: _____ Date _____
(Printed Name/ Signature)

Eric B. May

Ryn Gedra

Representative of the Graduate Dean (Dissertation Only):
(Printed Name/ Signature)

(To be returned to the Graduate School upon completion)

Salinity Intrusion Effects on Loblolly Pine (*Pinus taeda L.*) along the Delmarva Peninsula: an
analysis of wood density and morphology

by

Samantha Jalkowski

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Thesis Committee

Dr. Stephanie Stotts, Chair

Dr. Eric May

Dr. Keryn Gedan

Abstract

Ghost forests are striking indicators of salinization along the Delmarva peninsula. Despite increasing visibility, awareness, and research regarding coastal forest response to salinization, the mechanisms associated with tree decline and mortality are still not well understood, leaving an important gap with regards to conservation and restoration. This study seeks to elucidate wood density response to salinization by collecting and analyzing wood density measurements from *Pinus taeda* at Milford Neck, DE, Monie Bay, MD, and Brownsville Forest, VA with a Resistograph. Low forest samples captured tree response in a highly salt impacted system with elevated soil salinity and obvious tree mortality while high forest samples at the same location reflected minimal salt exposure. Wood density analysis revealed that salt exposed trees exhibited lower wood density than the high forest trees at all locations despite similar ages and density profiles revealed low wood density production near the pith and bark. A cellular comparison, including lumen area and cell wall thickness, was completed to identify the underlying source of density differences. Thin sections were collected for microscopic analysis using a digital image system and compared with statistical analysis. While the lumen area was similar at all sites, salt exposed trees at Monie Bay had thinner cell walls throughout the entire growth ring, but differences in cell wall thickness were restricted to earlywood at Milford Neck and latewood at Brownsville. This study provides important insights regarding wood density and cellular response of *P. taeda* to salinization and highlights the need for further investigation.

Table of Contents

Abstract	ii
List of Tables	iv
List of Figures	v
Introduction	1
2. Materials and Methods	7
2.1 Study Site and Experimental Design	7
2.2 Wood Density Sample Collection and Processing	9
2.3 Core Sampling and Wood Cell Processing	12
2.4 Tree Age Analysis	15
3. Results	16
3.1 Diameter and Age	16
3.2 Resistograph Wood Density	17
3.3 Core Density Profile	20
3.4 Wood Cellular Response to SLR: Cell Wall and Lumen Measurements	22
4. Discussion	27
5. Conclusion	32
Supplementary materials:	33
Funding: This research was made possible by the National Science Foundation Critical Zone Network Cluster Grant No. 2012484	33
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References	34

List of Tables

Table 1. Study Site Description. Soil salinity, elevation (NAVD88), and light availability data from Messerschmidt et al. (2022).	9
Table 2. Results Summary. Mean results for each site and elevation are reported with standard errors.....	16

List of Figures

Figure 1. (A) A map of lateral forest retreat rates (resolution 0.075 km ²) shows variability through the mid-Atlantic region. Areas of accelerated retreat at the Blackwater National Wildlife Refuge are highlighted with a white box. (B) The size of the white circles reflects relative sea-level rise rate (RSLRR) recorded by long-term tide gauges overlaid on watershed scale average lateral forest retreat rates. (C) Lower elevations show higher forest retreat rates. (D) Lateral forest retreat rates are not well correlated with RSLRR. (figure from Chen and Kirwan, 2023).	2
Figure 2. The hypothesis in this study was based, in part, on the diagram published by Cuny et al. (2014). According to this model, salinity stress (right) would cause smaller cells and thicker cell walls, leading to higher density wood.....	6
Table 1. Study Site Description. Soil salinity, elevation (NAVD88), and light availability data from Messerschmidt et al. (2021).	8
Figure 3. This study includes three locations on the Delmarva peninsula. Milford Neck (green star) is in the Delaware Bay watershed; Monie Bay (blue star) is in the Chesapeake Bay Watershed; and Brownsville (red star) drains to the coastal lagoons of the Atlantic Ocean.....	9
Figure 4. Trees were sampled with a Resistograph from the high forest (green dots) and low forest (blue dots) at the project study sites including, A) Milford Neck WMA, Delaware, B) Monie Bay NERR, Maryland, and C) Brownsville Preserve, Virginia. Yellow and red triangles represent trees that were both measured with the Resistograph and randomly selected for core analysis.....	11
Figure 5. (A) Raw data was trimmed at the red lines to remove the pith and bark. B) After flipping and detrending the raw data, the final data was oriented from pith to bark with the effect of friction removed.	12
Figure 6. Cell measurements were collected from a full row to capture trends in both cell wall and cell diameter present over an annual cycle (Cuny et al 2014).....	13

Figure 7. The lumen area of all cells (indicated with yellow outlines) in a full row were measured Image j analysis.	14
Figure 8. Wood density (A), DBH (B), and tree age (C) measurements comparing high forest (striped) and low forest (solid) measurements with stars indicating significant differences at one site and diamonds indicating significant differences between sites and forests.	19
Figure 9. Core density profiles (0 on the x axis represents the pith of the cores) for (A) Brownsville, (B) Monie Bay, (C) Milford Neck. Blue lines represent the low forest, and orange lines represent the high forest. Data was smoothed with a 20 unit running mean.	21
Figure 10. Analysis of lumen area (A) and cell wall thickness (B) with significant differences between elevation groups indicated with a star and site groups indicated with a diamond.	25
Figure 11. Analysis of (A) lumen area of EW (earlywood), T (transition), and LW (latewood) for all sites and elevations and (B) cell wall thickness comparison of EW, T, LW for all sites allows for a comparison of high forest (stripes) and low forest (solid). Stars indicate statistically significant differences between cell EW, T, or LW.	26
Figure 12. The carbon death spiral in coastal forest systems is depicted with a mortality spiral triggered by saltwater exposure. Carbon starvation and hydraulic failure are driven by various environmental conditions (McDowell et al, 2022).	31

Introduction

Ghost forests, large areas of remnant dead trees, are late-stage indicators of saltwater intrusion (Smart et al., 2020) and serve as a striking and common visual indicator of the transition between migrating salt marsh and the upland forest boundary (Ury et al., 2021; Walters et al., 2021). Accelerated rates of sea level rise (SLR) in the Chesapeake Bay region range from 3 - 6 mm per year (Smith and Kirwan, 2021) and drive the creation of ghost forests. Over 100,000 acres of forested ecosystems have been lost in the Chesapeake Bay region since the last century (Schieder et al., 2018), and predicted increases in SLR will translate to the conversion of thousands of additional acres of upland forest to marsh across the United States in this century (Chen and Kirwan, 2024).

Lateral forest retreat rates in the mid-Atlantic region are variable with an average of 0.67 (± 0.01) m per year. Elevation influences forest retreat rates (figure 1C), and some areas show little to no lateral retreat due to protection from barrier islands (figure 1A and 1B). Forests on the Delmarva peninsula appear to exhibit resilience to rising seas as only 40% of the coastal forests with predicted retreat show evidence of active retreat. Furthermore, there is little correlation between forest retreat and current sea level rise (figure 1D), but the correlation is greatly improved when examined with a 50-year lag (Chen and Kirwan, 2023). While areas with low elevation and gentle slopes will be most affected by SLR and saltwater intrusion, more research is needed to explain the lag between salinity intrusion and forest mortality.

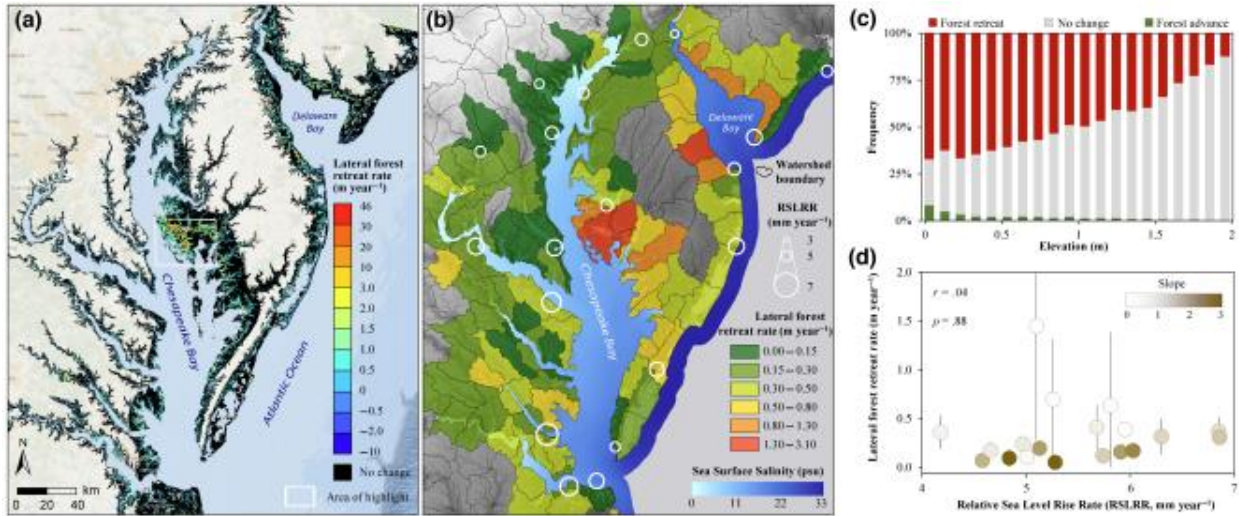


Figure 1. (A) A map of lateral forest retreat rates (resolution 0.075 km²) shows variability through the mid-Atlantic region. Areas of accelerated retreat at the Blackwater National Wildlife Refuge are highlighted with a white box. (B) The size of the white circles reflects relative sea-level rise rate (RSLRR) recorded by long-term tide gauges overlaid on watershed scale average lateral forest retreat rates. (C) Lower elevations show higher forest retreat rates. (D) Lateral forest retreat rates are not well correlated with RSLRR. (figure from Chen and Kirwan, 2023).

Reductions in coastal forest coverage are concerning as these systems are ecologically and economically important, providing coastal protection from storms (Van Loon Steensma and Vellinga, 2013; Van Dolah et al., 2020), a buffer for adjacent agricultural production (Sudol et al., 2023), protection for terrestrial and aquatic species, popular destinations for ecotourism (Mimura, 2013), and a foundation for commercial timber harvests and recreational hunting (Van Dolah et al., 2020).

As terrestrial forests fringing the upland marsh become waterlogged and salinized due to SLR, root oxygen levels, overall vigor, growth rates, crown coverage, carbon sequestration,

resilience against pests, and foliage size all decline, leading to increased mortality (Sudol et al., 2023). In the mid-Atlantic region, *Pinus taeda*, or loblolly pine, has shown some ability to tolerate low salinity conditions (Poulter et al., 2008) and may even benefit from decreased competition and increased light availability resulting from the mortality of more sensitive species (Wilkinson, 2018).

Pinus taeda is the dominant species in most mid-Atlantic forests subject to salinization and is the leading timber species in the south with over 13.4 million acres of coverage (Zhang et al., 2010). The species occurs from Florida to Delaware and as far west as Texas (Schultz, 1997). Significant areas of marginal agricultural land were converted to *P. taeda* plantations in the early 20th century as this species could thrive in well drained coastal soils (Zhang et al., 2010). Although many species compete with *P. taeda* for water, nutrients, and light availability, this species has an advantage along the marsh/forest boundary, even extending into the marsh (Schultz, 1997; Zhang et al., 2010).

Understanding the impact of salinization on *P. taeda* growth and vigor is important as plantations of this species in the southern U.S. remove more carbon from the atmosphere than any other terrestrial region in the country. The increase of *P. taeda* plantations in the southern U.S. and intense silviculture management practices have contributed to accelerated rates of atmospheric carbon removal. Over 747 Tg carbon is sequestered in above ground biomass within *P. taeda* and *Pinus echinata* forests (Aspinwall et al., 2012). However, when exposed to salinity stress, trees may become a carbon source rather than a sink (Aguilos et al., 2020).

Despite the growing spatial coverage of ghost forests and loss of economically and ecologically valuable ecosystem services, the driving processes and mechanisms associated with forest retreat are still not fully understood (Kirwan et al., 2025). While there has been increased

efforts to elucidate the processes and mechanisms associated with forest retreat (e.g., Messerschmidt et al., 2021; Chen and Kirwan, 2023; Norwood et al., 2021; Walters et al., 2021; Polle and Chen, 2015), research examining the effects of salinity stress on wood density, the dry mass for a given volume of wood, for non-mangrove systems has been minimal. The lack of research regarding salt stress is critical as wood density is the best predictor of wood quality (Fundova et al., 2018) because it can indicate strength, durability, and workability of the wood (Boakye et al., 2023).

There are numerous climate and non-climate related factors that can contribute to wood density, including genetics (Roth et al., 2007), age (Boakye et al., 2023), species (Fernandes et al., 2018; Roth et al., 2007), soil moisture (Camarero and Hevia, 2020), temperature (Gindl et al., 2000), pests (Dodds et al., 2018), competition (Poulter et al., 2008), topography (Ury et al., 2021; Boakye et al., 2023), light availability, and windstorms (Boakye et al., 2023). Restoration and management practices (Boakye et al., 2023; Roth et al., 2007), such as controlled burns (McNichol et al., 2019; Boakye et al., 2023), may also indirectly influence wood density. Soil salinity has been associated with increased wood density when salt induced stress causes plants to partition their biomass to strengthen their xylem to reduced the risk of cavitation (Stiller, 2009).

In addition to correlations with wood durability (Perrot et al., 2020; Schilling et al., 2021) and wood quality (Fundova et al., 2018), wood density has important ecological implications because it contributes to the hydraulic conductivity of trees (Boakye et al., 2023). Osmotic stress caused by saline environments or drought conditions can lead to the dehydration of plants, which affects cell cycles and cause xylem tension, which leads to cavitation (Cone, 2021; Muilu-Makela et al., 2015). Cavitation occurs when air bubbles form in the xylem, which typically

transport fresh water, leading to embolism and potential mortality if hydraulic conductivity is reduced to the point of water limitation (Ditsche and Summers, 2014). Little work has been done to examine the impact of non-drought salt stress on temperate forest species and the potential to cause cavitation.

While many studies report correlations between annual ring width and salinization (Fernandes et al., 2018; Candel-Perez et al., 2018; Hall et al., 2022), salinity exposure may also contribute to changes in wood density due to alterations in cell morphology, including cell size, wall thickness (Meier and Newton, 1987; Bjorklund et al., 2020; Cuny et al., 2014), or lignification (Bjorklund et al., 2020).

This study combines Resistograph measurements and cellular analysis of *P. taeda* in salinizing stands to examine the impact of inundation on wood density and cell morphology. We hypothesize that trees growing at lower elevations and higher salinity exposure will exhibit higher wood density and smaller cell size with thicker cell walls, resembling trees in arid environments (figure 2). The results of this study provide important insights regarding wood density and morphology response of *P. taeda*, the dominant tree species in the region, to salinization which will better inform system modeling through an increased understanding of forest transition, carbon storage, and hydrodynamics.

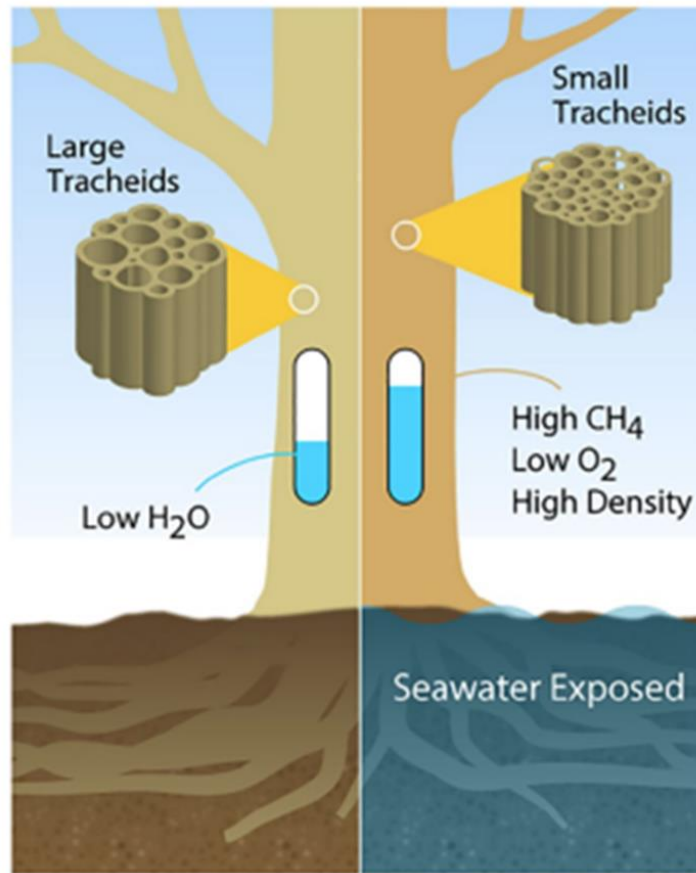


Figure 2. The hypothesis in this study was based, in part, on the diagram published by Cuny et al. (2014). According to this model, salinity stress (right) would cause smaller cells and thicker cell walls, leading to higher density wood.

2. Materials and Methods

2.1 Study Site and Experimental Design

This study includes data collection from all three states of the Delmarva peninsula, located in the mid-Atlantic region (figure 3). Study sites include the Milford Neck Wildlife Management Area (WMA; 38°58'5.3256"N, 75°22'11.5392"W) managed by Delaware Fish and Wildlife, Monie Bay (38°12'42.894"N, 75°48'39.1644"W) managed by Maryland National Estuarine Research Reserve (NERR), and the Nature Conservancy's Brownsville Preserve (37°27'48.2868"N, 75°50'5.5788"W; table 1 and figure 3). Brownsville forest surrounds Phillips Creek, which has been used as the site name in some published studies. All three sites were established by the Coastal Critical Zone Network (CCZ).

The climate in the southern mid-Atlantic region generally has warmer temperatures, especially as elevation drops in the coastal plain (Butler-Leopold et al., 2018). The Chesapeake Bay, Delaware Bay, and coastal lagoons on the eastern coast of Delmarva contain a large series of barrier islands, extensive saltmarsh ecosystems, and major large riverine estuaries. The mixed forest in the coastal plain of the mid-Atlantic has a distinct maritime climate including high humidity, mild winters, hot summers, rare droughts, and abundant precipitation (Butler-Leopold et al., 2018).

The CCZ project includes transects extending along a salinity gradient from the marsh to the high forest at all sites, in a space-for-time design. Data in this study was collected from the low and high forest zones. The low forest zones exhibit lower elevation, higher soil salinity, greater presence of marsh species in the understory (e.g., *Phragmites australis* and *Spartina patens*), lower woody plant diversity, and a more open canopy than the high forest (table 1).

There is more light availability in the low forest and less canopy competition (Nordio et al., 2024), however trees found in the low forest are more prone to flooding.

Pinus taeda (loblolly pine) and *Morella cerifera* (wax myrtle) were common to all study sites and elevations included in this project with *P. taeda* as the dominant species. Other plants commonly found in the high forest canopy included *Acer rubrum* (red maple), *Ilex opaca* (American holly), *Liquidambar styraciflua* (sweet gum), and *Nyssa sylvatica* (black gum). Plant species commonly found in the low forest included *Baccharis halimifolia* (salt bush), *Phragmites australis* (common reed), and *Toxicodendron radicans* (poison ivy; Messerschmidt et al., 2022).

Table 1. Study Site Description. Soil salinity, elevation (NAVD88), and light availability data from Messerschmidt et al. (2021).

Sites	Dominant Tree Species	Geographic Location	Elevation (m)		Adjacent Water	Average Soil Salinity (ppt)		Light Availability (% Density Coverage)	
			High	Low		High	Low	High	Low
Milford Neck WMA, DE	<i>Pinus taeda</i>	38°58'5.3256"N 75°22'11.5392"W	3.10	1.29	Delaware Bay	0.06	1.09	99.4	59.4
Monie Bay NERR, MD	<i>Pinus taeda</i>	38°12'42.894"N 75°48'39.1644"W	0.99	0.63	Chesapeake Bay	0.33	7.36	92.3	19.0
Brownsville Preserve, VA	<i>Pinus taeda</i>	37°27'48.2868"N 75°50'5.5788"W	1.02	0.94	Atlantic Ocean	0.56	10.30	95.4	34.9

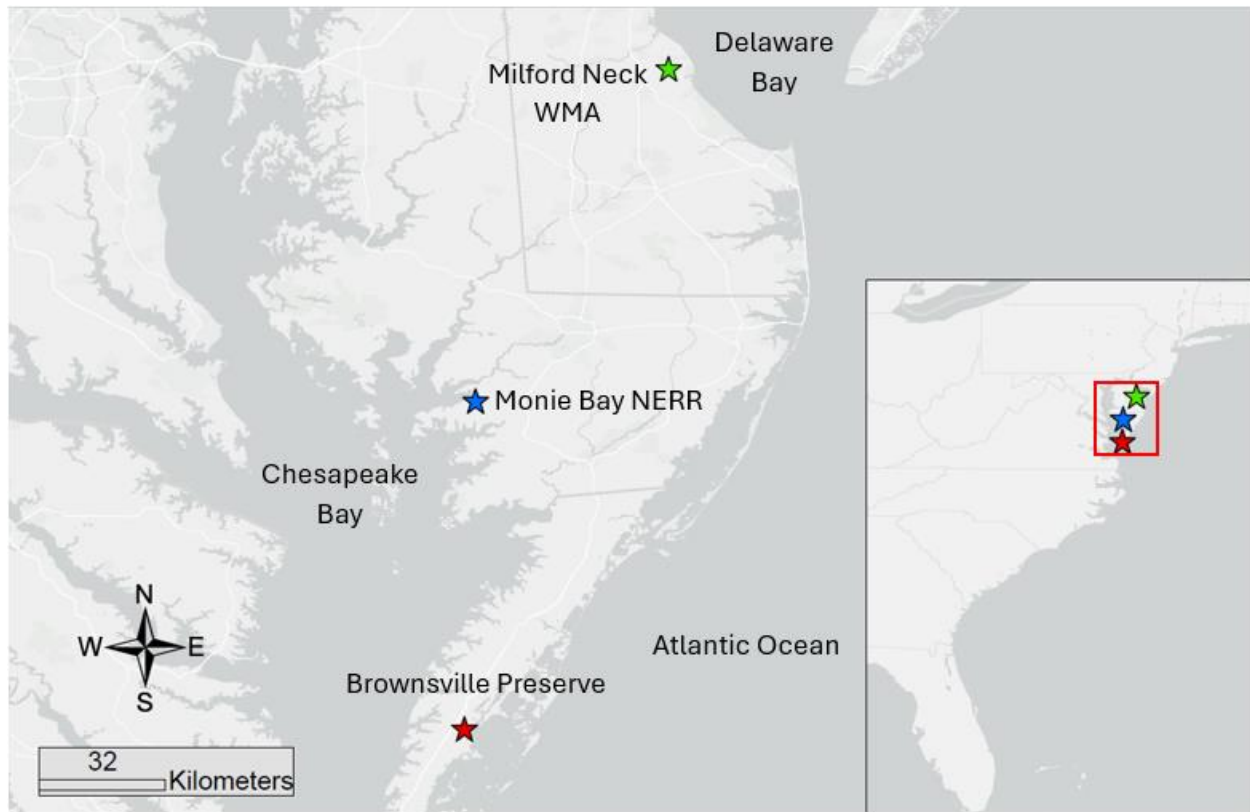


Figure 3. *This study includes three locations on the Delmarva peninsula. Milford Neck (green star) is in the Delaware Bay watershed; Monie Bay (blue star) is in the Chesapeake Bay Watershed; and Brownsville (red star) drains to the coastal lagoons of the Atlantic Ocean.*

2.2 Wood Density Sample Collection and Processing

A Rinntech 650 Resistograph estimates wood density by measuring the amplitude of resistance encountered when drilling into wood (Isik and Li, 2003). Resistograph readings were collected at breast height (1.2 meters from base of the tree) from all live *P. taeda* with a diameter at breast height (DBH) greater than 12.7 cm within each study site. Using the Resistograph phone application, the Latitude, Longitude (figure 4), and density measurements of each sampled tree were recorded after the tree was drilled and tagged.

Data from the Resistograph was downloaded into the DECOM program where it was edited before analysis (figure 5A). Resistograph output was trimmed at the bark and pith, flipped so that the pith was oriented on the left, and detrended linearly (figure 5B; Fundova et al., 2018). Using the R studio Densitr package (2024.04.2) and Microsoft Excel, Kruskal-Wallis rank sum test and pairwise Wilcox test were used to analyze wood density. The DBH was extracted from the Resistograph measurements for all trees sampled, and the data was analyzed with Kruskal-Wallis rank sum test and pairwise Wilcox test.

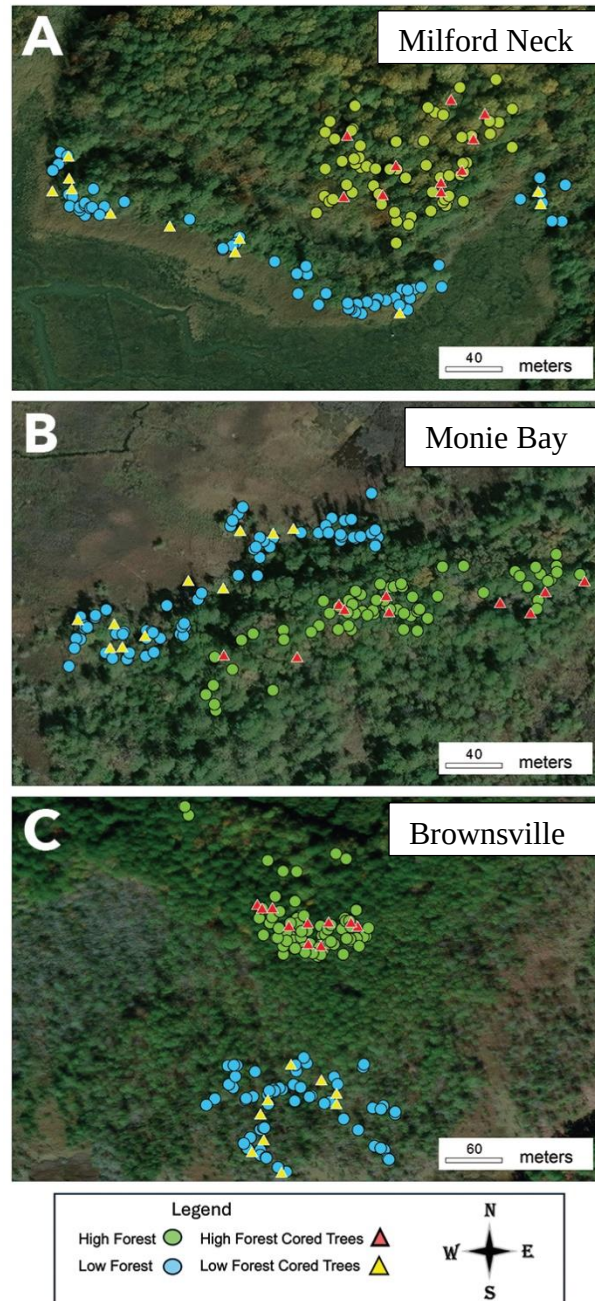


Figure 4. Trees were sampled with a Resistograph from the high forest (green dots) and low forest (blue dots) at the project study sites including, A) Milford Neck WMA, Delaware, B) Monie Bay NERR, Maryland, and C) Brownsville Preserve, Virginia. Yellow and red triangles represent trees that were both measured with the Resistograph and randomly selected for core analysis.

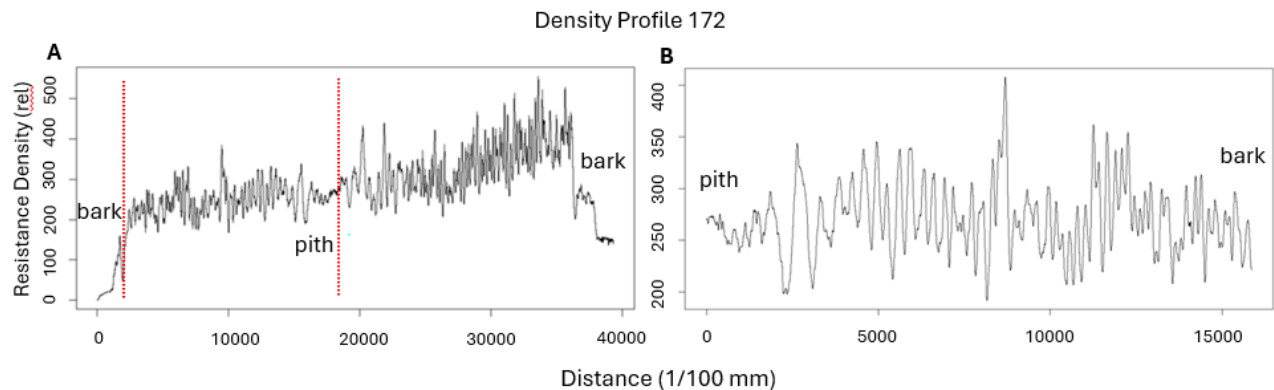


Figure 5. (A) Raw data was trimmed at the red lines to remove the pith and bark. B) After flipping and detrending the raw data, the final data was oriented from pith to bark with the effect of friction removed.

2.3 Core Sampling and Wood Cell Processing

Ten high forest trees and ten low forest trees were selected for coring and cell analysis using stratified random design and random number generator. Cores were collected from the selected trees at breast height with a 5 mm increment borer and were dried for 24-48 hours before gluing them to a wooden mount.

Annual rings were selected for cellular analysis based on their average wood density measurement, which was estimated by averaging consecutive peak and valley Resistograph readings. For each core, rings corresponding to the minimum, 1st quartile (25th percentile), median (50th percentile), 3rd quartile (75th percentile), and maximum wood density measurements were identified with the R studio Densitr package (2024.04.2) and excel for thin sectioning and analysis. Slices 2-10 μ m thick were collected from the selected annual rings with a microtome-GSL. Slides of the thin sectioned samples were created by dyeing the sections with Safranin:Astra

blue mixture and mounting with Canada Balsam and Glycerol (Gartner and Schweingruber, 2013).

Images of the earlywood, transition, and latewood cells (figure 6) at x1000 magnification were collected with a Ziess Zen blue camera. Both the lumen area (figure 7) and cell wall thickness were measured with Image J from one full row of cells with row selection based on picture quality and ring visibility (Cuny et al., 2014). Both the total cell area and porosity were calculated from the cell measurements. Porosity, which measures the open space within the cells, was estimated by dividing the average lumen area by average total cell area (Acquah et al., 2018).

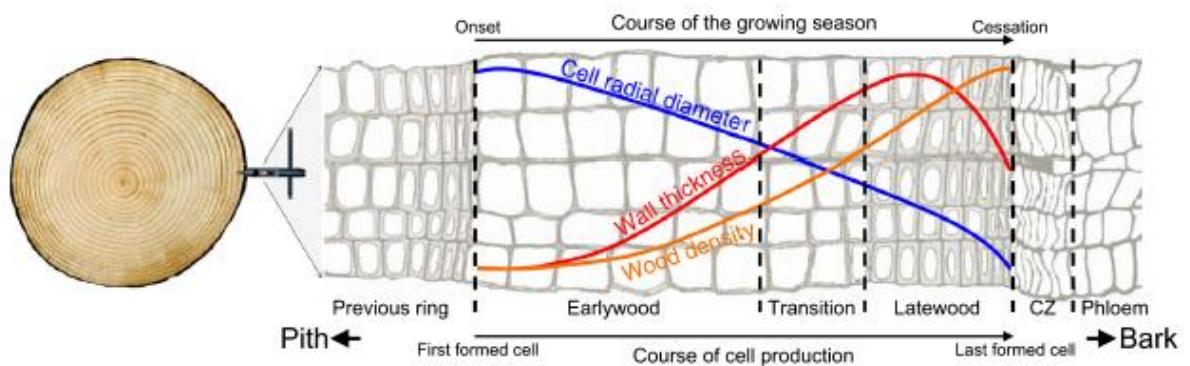


Figure 6. Cell measurements were collected from a full row to capture trends in both cell wall and cell diameter present over an annual cycle (Cuny et al 2014).

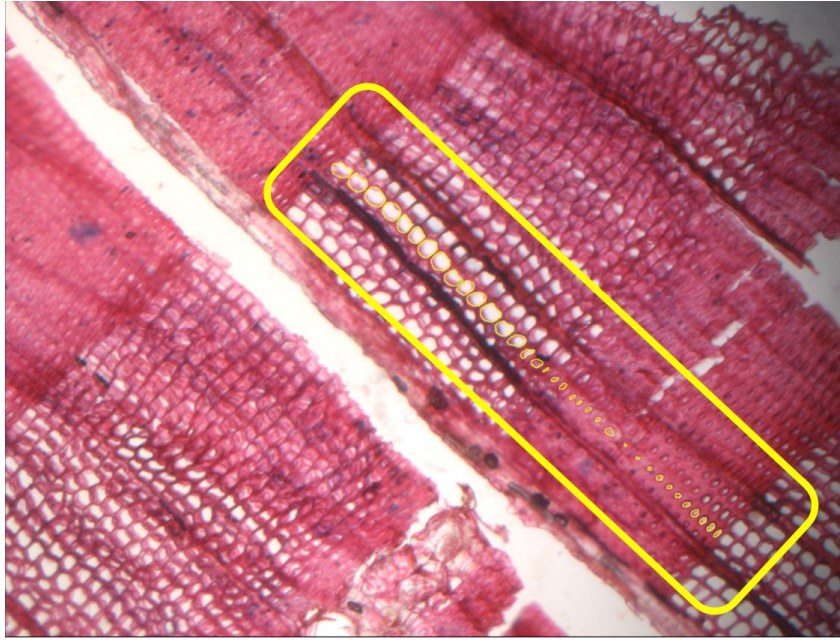


Figure 7. The lumen area of all cells (indicated with yellow outlines) in a full row were measured Image j analysis.

Cell measurements were compared between site (Brownsville, Monie Bay, Milford Neck), elevation (high or low forest), and distribution (Minimum, 1st quartile, median, 3rd quartile, maximum) with non-parametric tests in R (2024.04.2), including the Kruskal-Wallis Rank sum test, pairwise Wilcox test, and PERMANOVA using a 0.05 alpha level.

Z scores were used to standardize data and visualize density trends over the core length. The distance and average amplitude (density) between each peak and valley from the processed Resistograph data were recorded. Distances between peaks and valleys were converted to percent length to allow for comparison of different sizes and ages of trees. Density z scores were calculated with the equation $Z = ((\text{average amplitude of the peak and valley}) - \text{mean}) / \text{standard deviation}$ for each core (Bjorklund et al., 2019). Group average z scores were then plotted with a 20 unit moving average for visual interpretation.

2.4 Tree Age Analysis

After thin sectioning, mounted and dried cores were sanded with progressively finer sandpaper coarseness (220 to at least 800). Tree rings were counted and measured using a Velmex linear encoder with 0.001 mm precision, and annual ring measurements were cross dated with COFECHA (Holmes, 1983). Non-parametric tests in R (2024.04.2) were used for the statistical analysis, including the Kruskal-Wallis Rank sum test and pairwise Wilcox test..

3. Results

Measurements were collected from a total of 451 trees, and 40,603 cells from 144 rings (Table 2).

Table 2. Results Summary. Mean results for each site and elevation are reported with standard errors.

	Milford Neck		Brownsville		Monie Bay	
	High	Low	High	Low	High	Low
Mean DBH (cm)	49.25 (± 1.36)	23.81 (± 1.57)	35.27 (± 1.01)	25.34 (± 1.00)	36.61 (± 1.21)	27.88 (± 1.06)
Mean Wood Density (Amplitudes)	570.36 (± 16.10)	394.65 (± 18.58)	536.18 (± 12.97)	377.76 (± 8.10)	523.59 (± 18.91)	409.92 (± 13.26)
Mean Tree Age (Years)	80.30 (± 5.07)	55.09 (± 6.43)	74.47 (± 4.57)	81.50 (± 2.19)	75.10 (± 3.66)	74.47 (± 3.06)
Mean Lumen Area (μm^2)	30049.00 (± 1570.34)	28080.71 (± 1243.46)	25471.52 (± 1712.34)	24701.12 (± 1286.91)	25237.08 (± 1537.16)	24597.49 (± 1343.64)
Mean Cell Wall Thickness (μm)	31.14 (± 0.62)	29.67 (± 0.71)	31.04 (± 0.63)	30.04 (± 0.89)	34.45 (± 1.06)	28.45 (± 0.67)
Mean Cell Area (μm^2)	51893.04 (± 2068.99)	48244.31 (± 1565.42)	45812.73 (± 2374.55)	44669.47 (± 1976.23)	47834.87 (± 1992.85)	42997.18 (± 1789.53)
Mean Porosity (%)	56.58 (± 1.01)	57.42 (± 0.95)	54.17 (± 0.91)	54.54 (± 1.01)	51.24 (± 1.39)	56.01 (± 1.04)

3.1 Diameter and Age

The DBH at Milford Neck ranged from 8.89-80.63 cm with a high forest mean of 49.25 (± 1.36) cm and a low forest mean of 23.81 (± 1.57) cm ($p < 0.001$; figure 8B). Tree diameters at

Milford Neck high forest were significantly greater than Monie Bay high forest (mean = 36.61 $\pm$ 1.21 cm, $p < 0.001$) and Brownsville high forest (mean = 35.27 $\pm$ 1.01 cm, $p < 0.001$). Milford Neck low forest trees also had a significantly different DBH than those at Monie Bay low forest (mean = 27.88 $\pm$ 1.06 cm; $p < 0.05$; table 2).

Monie Bay DBH ranged from 7.97-64.39 cm and Brownsville ranged from 8.32-60.02 cm. Both sites showed significant differences between the high and low forest DBH. Monie Bay high forest had a mean DBH of 36.61 ($\pm$ 1.21) cm, and the low forest mean DBH was 27.88 ($\pm$ 1.06) cm (Table 2; figure 8B; $p < 0.001$). Brownsville also exhibited a significant difference ($p < 0.001$) between the high forest (mean DBH = 35.27 ($\pm$ 1.01) cm) and the low forest (mean DBH = 25.32 ($\pm$ 1.00) cm; table 2; figure 8B).

Milford Neck tree age ranged from 10–92 years old. Milford Neck high forest (mean age = 80.30 $\pm$ 5.07 years; table 2) and Milford Neck low forest (55.09 $\pm$ 6.43 years; table 2; figure 8B) were significantly different in age ($p < 0.01$). Brownsville low forest (81.50 $\pm$ 2.19 years) was significantly older than Milford Neck low forest with a mean age of 55.09 $\pm$ 6.43 years ($p < 0.05$).

Monie Bay tree age ranged from 47–107 years old and there was no significant difference between the age of high and low forest trees at that site (figure 8C). Brownsville tree age ranged from 60-138 years old and showed no significant difference in age between the high and low forests at that site (figure 8C).

3.2 Resistograph Wood Density

Mean resistance measurements, reflecting wood density, ranged from 336.18 ($\pm$ 12.97) amplitudes at Brownsville high forest to 570 ($\pm$ 16.10) amplitudes at Milford Neck high forest (table 2; figure 8A). Wood density at Milford Neck ranged from 960.90-197.47 amplitudes. Milford Neck high forest had a mean amplitude of 570 ($\pm$ 16.10) which was significantly

different ($p < 0.001$) than Milford Neck low forest with a mean amplitude of 394 (± 18.58) (table 2; figure 8A). Wood density at Monie bay ranged from 938.70-238.30 amplitudes. Monie Bay high forest had a mean amplitude of 523.59 (± 18.91) which was significantly different ($p < 0.001$) than Monie Bay low forest with a mean amplitude of 409.92 (± 13.26) (table 2; figure 8A). Brownsville wood density ranged from 880.10-229.60 amplitudes. Brownsville high forest had a mean amplitude of 536.18 (± 12.97) which was statistically different than Brownsville low forest with a mean amplitude of 377.76 (± 8.10) (table 2; figure 8A). There was no significant difference between the high or low forests when conducting site to site comparison (figure 8A).

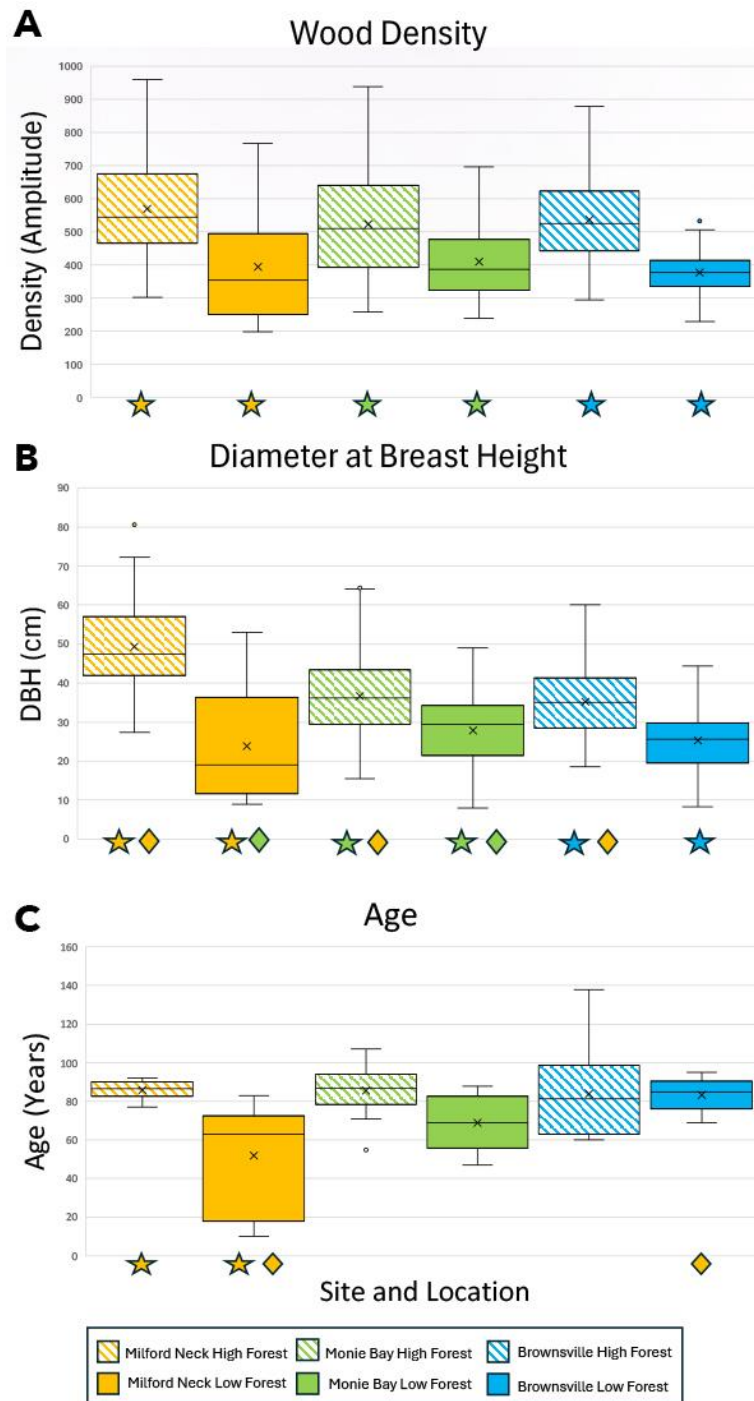


Figure 8. Wood density (A), DBH (B), and tree age (C) measurements comparing high forest (striped) and low forest (solid) measurements with stars indicating significant differences at one site and diamonds indicating significant differences between sites and forests.

3.3 Core Density Profile

Z scores ranged between 1.5 and -1.5 . At the Brownsville high and low forests, the z scores were negative for the first 25% and last 20% of the core length (figure 9A). Monie Bay high forest had a positive z score for the first 28% and last 15% of the core length (figure 9B), and Monie Bay low forest had negative z score for the first 37%, between 60-80%, and 87-100% of the core length (figure 9B). Milford Neck high forest had a negative z score for the first 27% and last 27% of the core length, and the low forest had a negative z score for the first 25% and last 30% of the core length (figure 9C).

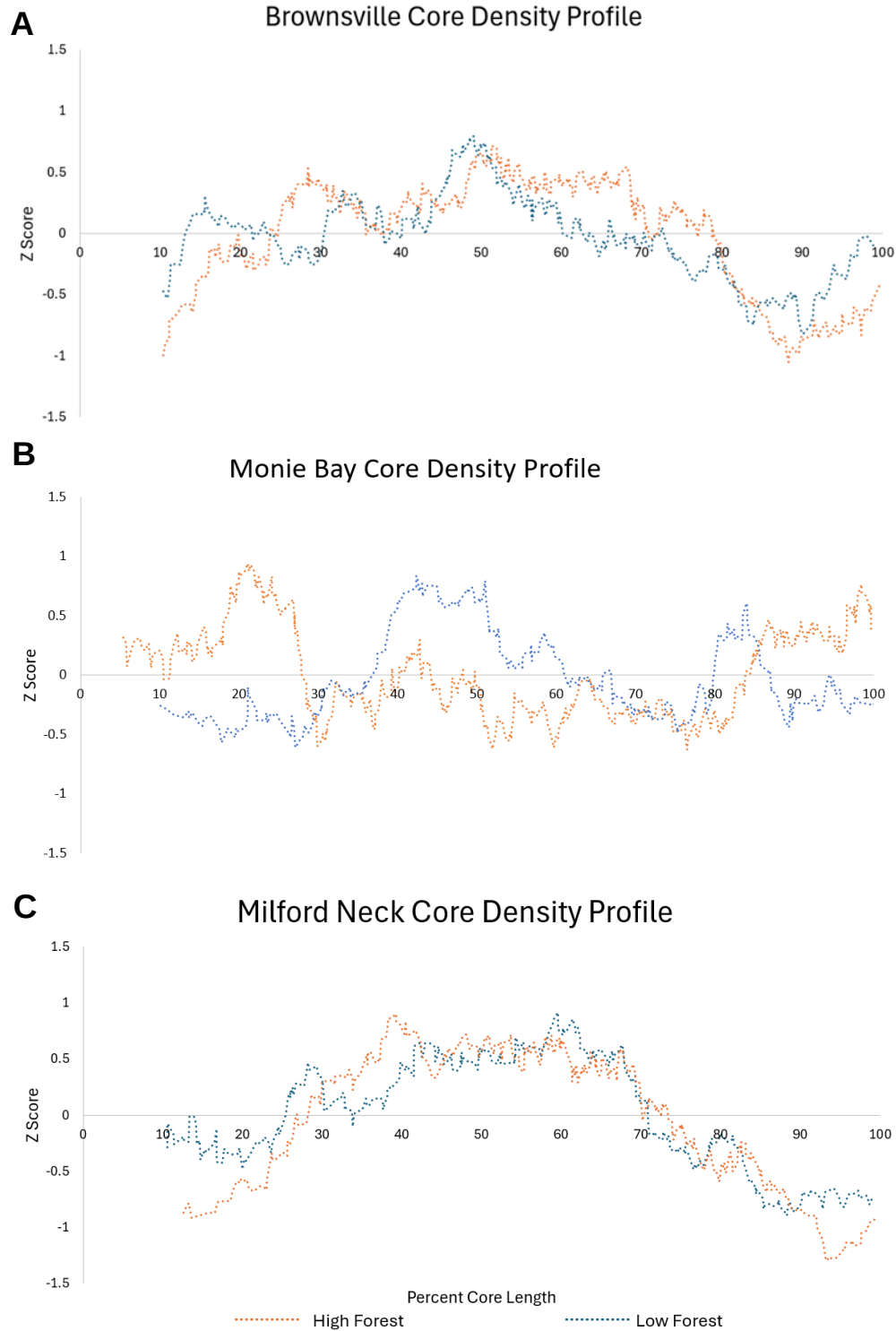


Figure 9. Core density profiles (0 on the x axis represents the pith of the cores) for (A) Brownsville, (B) Monie Bay, (C) Milford Neck. Blue lines represent the low forest, and orange lines represent the high forest. Data was smoothed with a 20 unit running mean.

3.4 Wood Cellular Response to SLR: Cell Wall and Lumen Measurements

Milford Neck lumen area ranged from 8332.95-55528.2 μm^2 . The mean lumen area was 30049 (± 1570.35) μm^2 in the high forest and 28080.70 (± 1243.47) μm^2 in the low forest (table 2). Monie Bay lumen area ranged from 7118.73-53321.2 μm^2 . Monie Bay high forest had a mean lumen area of 25237.10 (± 1537.16) μm^2 , and the low forest had a mean lumen area of 24597.5 (± 1343.64) μm^2 (table 2). Brownsville lumen area ranged from 4648.38-66074.60 μm^2 . Brownsville high forest had a mean lumen area of 25471.50 (± 1712.35) μm^2 , and the low forest had a mean lumen area of 24701.10 (± 1286.91) μm^2 (table 2). No site or forest showed any significant difference in lumen area (figure 10A).

The mean lumen area at Brownsville high forest was 40198.07 (± 2634.80) μm^2 in the earlywood, 29013.98 (± 2445.41) μm^2 in the transition, and 7202.51 (± 462.09) μm^2 in the latewood. At Brownsville low forest the mean lumen area in the earlywood was 42424.96 (± 2498.91) μm^2 , 25618.29 (± 1703.63) μm^2 in the transition, and 6060.11 (± 412.51) μm^2 in the latewood.

Monie Bay high forest had a mean lumen area of 44103.08 (± 2918.93) μm^2 in the earlywood cells, 23422.24 (± 1864.05) μm^2 in the transition, and 8185.90 (± 582.14) μm^2 in the latewood. Monie Bay low forest had a mean lumen area of 42204.63 (± 2633.71) μm^2 in the earlywood cells, 24363.46 (± 1869.10) μm^2 in the transition, and 7224.367 (± 596.18) μm^2 in the latewood. Milford Neck high forest had a lumen area of 50800.05 (± 3010.03) μm^2 in the earlywood cells, 29257.67 (± 1992.24) μm^2 in the transition cells, and 10089.29 (± 743.65) μm^2 in the latewood cells. Milford neck low forest had a mean lumen area of 46131.36 (± 2000.70) μm^2 in the earlywood cells, 29451.02 (± 2063.80) μm^2 in the transition cells, and 8659.74 ($\pm$

794.18) μm^2 in the latewood cells. Results showed no significant difference in lumen area between the high and low forest at each site (figure 11A).

Milford Neck cell wall thickness ranged from 19.42-50.26 μm . Milford Neck high forest had a mean cell wall thickness of 31.14 (± 0.62) μm , and the low forest had a mean cell wall thickness of 29.67 (± 0.71) μm (table 2). Brownsville cell wall thickness ranged from 19.41-49.93 μm . Brownsville high forest had a mean cell wall thickness of 31.04 (± 0.63) μm , and the low forest had a cell wall thickness of 30.67 (± 0.89) μm (table 2). Both Milford Neck and Brownsville showed no significant difference in cell wall thickness between the high and low forests (figure 10B).

Monie Bay cell wall thickness ranged from 17.58-55.72 μm . Monie Bay high forest had a mean cell wall thickness of 34.45 (± 1.06) μm , and the low forest had a mean cell wall thickness of 28.86 (± 0.67) μm (table 2). Monie Bay high and low forest cell wall thickness was statistically different ($p < 0.01$; Figure 10B). Monie Bay high forest was also statistically different than Brownsville low forest ($p < 0.05$; figure 10B).

At Brownsville high forest the mean cell wall thickness in the earlywood was 20.28 (± 0.56) μm , the transition was 25.91 (± 1.02) μm , and the latewood was 47.05 (± 1.09) μm . Brownsville low forest had an earlywood mean of 20.07 (± 0.80) μm , transition mean of 28.74 (± 1.33) μm , and latewood mean of 43.20 (± 1.26) μm . Brownsville site showed a significant difference in cell wall thickness in the latewood ($p < 0.05$, figure 11B).

Monie bay high forest earlywood had a mean cell wall thickness of 25.88 (± 1.14) μm , and the earlywood cell wall thickness in the low forest was 20.52 (± 0.62) μm . The transition cell wall thickness in the high forest had a mean of 33.90 (± 1.62) μm , and the low forest had a mean of 26.58 (± 0.91) μm . The latewood mean cell wall thickness in the high forest was 43.58 ($\pm$

1.28) μm , and in the low forest the mean was 39.48 (± 1.18) μm . Monie bay showed significant differences in the earlywood ($p < 0.0001$), transition ($p < 0.001$), and latewood ($p < 0.05$) cells between the high and low forests (figure 11B).

At Milford Neck high forest the mean cell wall thickness for the earlywood was 19.80 (± 0.65) μm . The transition mean cell wall thickness was 29.00 (± 1.20) μm , and the latewood cell wall thickness was 44.61 (± 0.85) μm . Milford Neck low forest had a mean cell wall thickness of 17.11 (± 0.38) μm in the earlywood, 27.80 (± 1.60) μm in the transition cells, and 44.10 (± 1.14) μm in the latewood cells. Cell wall thickness showed a significant difference at Milford neck earlywood between the high and low forest ($p < 0.001$, figure 11B).

Porosity at Milford Neck ranged from 35.11-74.41 %. Milford Neck high forest had a mean porosity of 56.58 (± 1.01) %, and the low forest mean porosity was 57.42 (± 0.95) % (table 2). Monie Bay porosity ranged from 30.87-69.46 %. Monie Bay high forest had a mean porosity of 51.24 (± 1.39) %, and the low forest had a mean porosity of 56.01 (± 1.04) % (table 2). Porosity at Brownsville ranged from 27.64-71.04 %. Brownsville high forest had a mean porosity of 54.17 (± 0.91) %, and the low forest had a mean porosity of 54.54 (± 1.10) %. There was no significant difference in porosity between forests at each site. Monie Bay high forest and Milford Neck low forest showed a significant difference in porosity ($p < 0.05$).

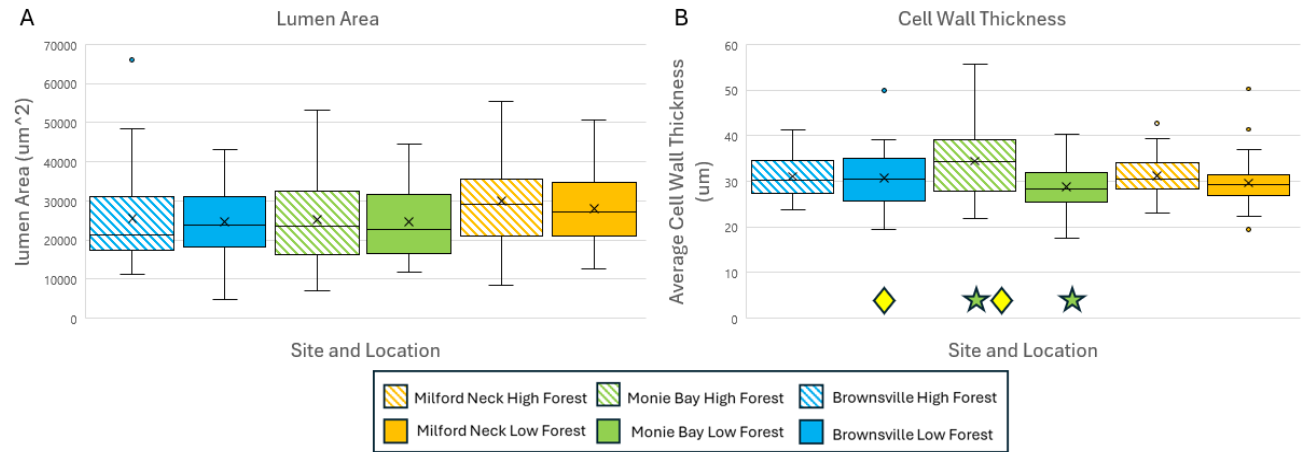


Figure 10. Analysis of lumen area (A) and cell wall thickness (B) with significant differences between elevation groups indicated with a star and site groups indicated with a diamond.

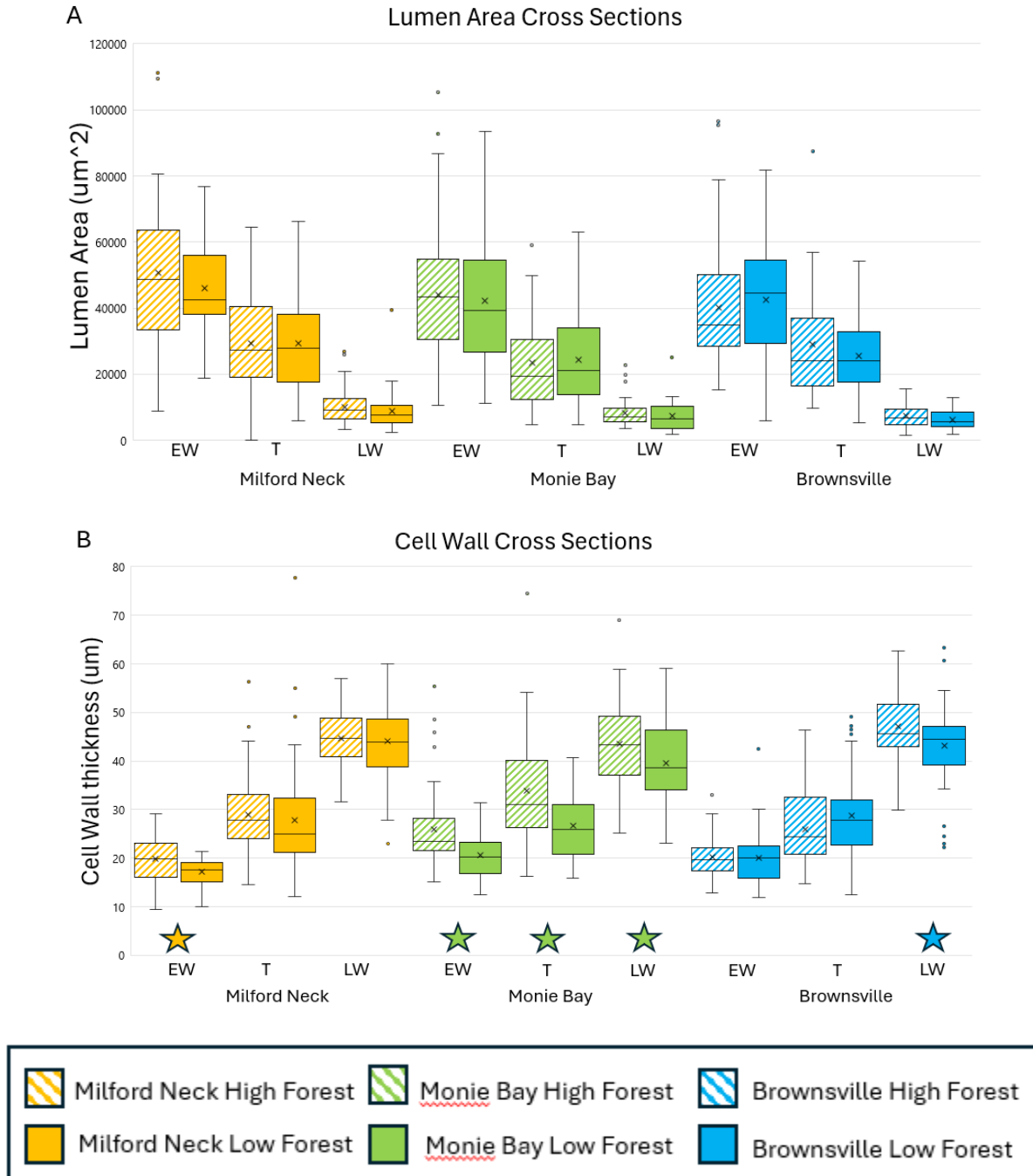


Figure 11. Analysis of (A) lumen area of EW (earlywood), T (transition), and LW (latewood) for all sites and elevations and (B) cell wall thickness comparison of EW, T, LW for all sites allows for a comparison of high forest (stripes) and low forest (solid). Stars indicate statistically significant differences between cell EW, T, or LW.

4. Discussion

The results of this study contradicted our initial hypothesis, as the mean wood density for salt exposed, low forest trees was less than that of the high forest trees at all sites, highlighting the complexity of wood density in salinizing coastal systems (figure 8A).

An analysis of *P. taeda* size and age established a basis for cross-site comparison of wood density and cell morphology. Strong similarity was observed among sites with some exceptions at Milford Neck. Tree age was mostly uniform across sites with differences isolated to Brownsville and Milford Neck low forests (figure 8C). Tree size, based on mean DBH, was also similar. Milford Neck high forest trees were larger than the other sites, and the low forest trees were smaller than those in the Brownsville low forest (figure 8B).

After comparing tree sizes between study sites, mean DBH was analyzed to identify size differences between elevation groupings at each site. We expected low forest trees to be smaller than high forest trees because they are subject to greater salt exposure, and studies have found reduced wood growth associated with elevated soil salinity (e.g., Naskar and Palit, 2015; Fernandez et al, 2018; Candel-Perez et al., 2018; Roth et al., 2007) and low elevation (Tucker, 2018). Data confirmed this hypothesis as the mean DBH in the low forest was significantly less than the high forest at all sites (figure 8B).

To determine if size differences between elevation groups were the result of suppressed growth rates or differences in age, we completed an age comparison between elevation groups at each site. The results revealed that the high and low forests were the same age at both Brownsville and Monie Bay but not at Milford Neck where trees in the low forest were younger than those in the high forest (figure 8C). Despite attempts at using the Resistograph to determine tree age, this method was unable to accurately distinguish between extremely small rings

common in salt exposed trees or young rings with minimal density differences between early and late wood. Therefore, age measurements were restricted to physical cores which could be manually measured and verified.

Low elevation trees exhibited significantly lower wood density than their high elevation counterparts at all sites despite any differences in age or size, but no differences in wood density were detected between sites (figure 8A). These results are somewhat surprising as they counter published hypotheses (Norwood et al., 2021; Cuny et al., 2014; Camarero and Hevia, 2020), but wood density is complex and studies of *Pinus* species in salinizing systems are rare (Loustau et al., 1994). The consistency in density measurements among sites is also interesting given that Milford Neck low forest had lower soil salinity than the other sites and less variability between the high and low forest (table 1).

In addition to salinity, other environmental factors may contribute to reduced wood density in the low forests. Surviving trees in declining coastal forests may experience increased light exposure when neighboring trees die and the crown canopy is reduced (Wilkinson, 2018). Accelerated growth rates associated with decreased competition could contribute to low wood density near the forest to marsh transition. Age effects have also been associated with reduced wood density in *P. taeda* with young wood exhibiting low density for the first 15-20 years of growth (Boakye et al., 2023; Roth et al., 2007). Age effects would only be relevant at Milford Neck, because the other sites were composed of similarly aged trees (figure 8C). Furthermore, efforts were made to reduce the influence of age by using large sample sizes and sampling over the wood density distribution.

Although wood density was averaged over the entire core length for site and elevation comparisons, the generation of standardized density profiles using Resistograph data allowed us

to explore and compare trends in wood density over the growth span of cored trees. Percent core length, rather than dates, were used due to difficulty identifying rings with the Resistograph and lack of processed tree ring chronologies. Periods of low-density growth were identified by z scores less than zero, and high-density growth was identified by z scores greater than zero.

Brownsville (figure 9A) and Milford Neck (figure 9C) showed similar density profile trends with low density growth occurring at the first and last of the core length for both the high and low forests. Low density wood production is typical in early tree growth (Boakye et al., 2023; Roth et al., 2007) and may be due to salinization effects in late growth at the low forest but is unexplained at the high forest. The Monie Bay low forest density profile showed similar trends as Brownsville and Milford Neck, but the high forest profile was converse to the other site with high density wood production in early and late growth, and low-density wood occurring mid-core (figure 9B).

Wood density is typically the product of cell size and cell wall thickness regardless of the environmental drivers (Cuny et al., 2014). Therefore, we hypothesized that the density differences observed in this study would correspond to cellular structure with lower densities explained by thinner cell walls or larger lumen areas. Understanding the cellular response to SLR could provide insights regarding tree adaptation to increased salinity and hydrologic conductivity as large diameter tracheids (lumen area) are more efficient at transporting water but make the tree more prone to cavitation, while smaller tracheids protect against cavitation but limit water transport capacity (Cuny et al., 2014). Cell wall thickness also affects tree resilience as thinner cell walls are associated with improved elasticity (Meier and Newton, 1987), protecting the tree against high winds common in coastal ecosystems (Ury et al., 2021; Fernandez et al., 2018).

However, reduced cell wall thickness may also increase vulnerability to cavitation (Cuny et al., 2014) and pests (Dodds et al., 2018).

Results of the cellular analysis identified differences in cell wall thickness, rather than cell size, associated with salinization. Analysis over both the full growth ring and the earlywood, transition, and latewood showed a consistent mean lumen area and porosity across all groupings (figure 10A and table 1). However, thinner cell walls were associated with the low forest at all sites. Differences in cell wall thickness averaged over a full growth ring were only detected at Monie Bay (figure 10B) but were found at all sites when analyzed by earlywood, transition, and latewood growth with reduced cell wall thickness in the low forest isolated to earlywood growth at Milford Neck and latewood growth at Brownsville (figure 10B).

Lignification, a process where plants increase or decrease their vascular body to provide structural support, is also known to increase wood density in stressful environmental conditions (Boakye et al., 2023). While lignification was not explored in this study, further analysis is underway to explore the impact of salinization on this parameter using samples from this project.

Reduced cell wall thickness, and potentially lignification, associated with salt exposure may be a response to diminishing carbon availability with important implications for climate resilience. The carbon death spiral (figure 12) has often been used to describe the decline and mortality of salt exposed trees who close their stomata to conserve water due to osmotic stress, limiting their photosynthetic rate and reducing carbon availability which eventually leads to starvation (Stiller et al., 2009). More research is needed to explain why the timing of reduced cell wall thickness varied between sites.

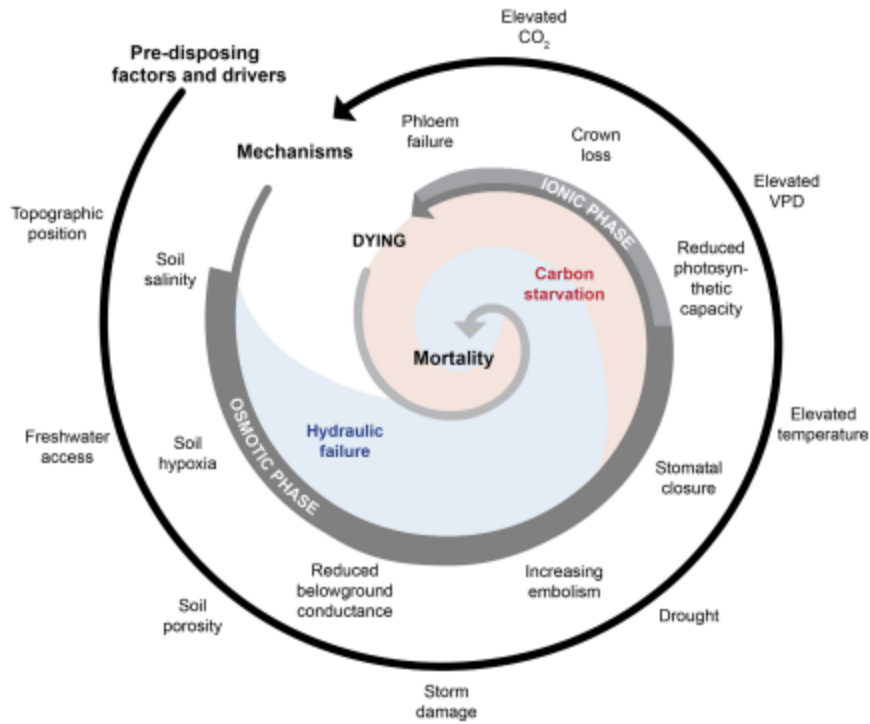


Figure 12. The carbon death spiral in coastal forest systems is depicted with a mortality spiral triggered by saltwater exposure. Carbon starvation and hydraulic failure are driven by various environmental conditions (McDowell et al, 2022).

The results of this study have important implications for climate resilience. *Pinus taeda* plays a key role in greenhouse gas reduction through carbon sequestration. The carbon stored in woody tissue of dead and decomposing trees enters the aboveground carbon pool when coastal forests retreat upland (Zhao et al., 2023). Furthermore, low density wood and reduced cell wall thickness associated with salinization indicate reduced carbon storage capacity of salt exposed trees. While methane production and storage were not examined in this project, increased methane release has been observed with salinization (Norwood et al., 2021; Seyfried et al., 2023). Quantifying the effect of upland forest retreat on greenhouse gas should be explored as a means of improving climate models.

5. Conclusion

This study provides a comparison of wood density and wood anatomy of *P. taeda* with differing salt exposure from three sites on the Delmarva peninsula. Trees growing in the low forests with greater salinity exposure displayed reduced growth compared to their upland counterparts with smaller diameters but similar age. Lower wood density measurements were found at the lower elevations with more salt exposure. The cellular analysis indicated that differences in wood density were associated with thinner cell walls, though the timing of reduced cell wall production was inconsistent.

Despite recent research into ghost forest development, there is a paucity of work focused on adult *P. taeda* wood density and cell morphology in salt stressed environments. Given the implications of morphological changes regarding hydrologic conductivity and resilience to wind, pests, and osmotic stress, changes associated with salinization and inundation should be further investigated.

The results of this study provide important insights regarding wood density and morphology response of *P. taeda* to salinization which will better inform system modeling through an increased understanding of forest transition, carbon storage, and hydrodynamics. This work's results are relevant to coastal ecologists and the commercial timber industry in coastal regions subject to SLR. *P. taeda* is the most common timber species on the Delmarva peninsula (Cope, 1921) and the most planted timber species in the southern U.S. (Schultz, 1997). While genetic modification and drought resistance in *P. taeda* have been subject to extensive research efforts, and few studies have focused on stomatal, root, and seedling growth in saline conditions, adult response to salinization is not well understood but crucial to understanding marsh transition and future land erosion.

Supplementary materials:

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